CODE-LOGIK
Crafting Cutting-Edge Software

THE WILLOW TREE OMS

CS4233 CAPSTONE PROJECT 2024

MODELING

PREPARED BY:

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PROJECT OVERVIEW

PROJECT NAME: The Willow Tree OMS
PROJECT REFERENCE: CS4233 CAPSTONE PROJECT 2024
DATE: 29 JAN 2024 – 29 APR 2024

OBJECTIVE: Redesign The Willow Tree's Order Management System (OMS) to create a user-friendly application that can keep up with menu changes and enhance the customer dining experience.

PROBLEM STATEMENT: The current OMS limits the menu items to those available during development and does not reflect the current menu. Because the menu has evolved, the need has arisen to redesign the OMS to reflect the current menu and allow future menu updates.

USER STORIES

USER STORY 1	
Story Title	View Menu
Story ID	001
Story Type	User
Functional Area	Design and Structure Menu Items and Selection
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	seated at a table
I want to…	be able to view the menu
In order to…	make selections

USER STORY 2	
Story Title	Select Menu Items
Story ID	002
Story Type	User
Functional Area	Design and Structure Menu Items and Selection
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	viewing the menu
I want to…	be able to select menu items
In order to…	prepare to place an order

USER STORY 3	
Story Title	Unselect Menu Items
Story ID	003
Story Type	User
Functional Area	Design and Structure Menu Items and Selection
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	viewing the menu
I want to…	be able to change menu item selections
In order to…	prepare to place an order

USER STORY 4	
Story Title	Cancel Order
Story ID	004
Story Type	User
Functional Area	Design and Structure Menu Items and Selection
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	viewing the menu
I want to…	be able to clear all menu selections
In order to…	cancel my order

USER STORY 5	
Story Title	View Order
Story ID	005
Story Type	User
Functional Area	Design and Structure Menu Items and Selection Order Summary
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	viewing the menu
I want to…	be able to view all menu selections
In order to…	ensure selection correctness

USER STORY 6	
Story Title	Place Order
Story ID	006
Story Type	User
Functional Area	Design and Structure Menu Items and Selection Order Summary
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	viewing the menu
I want to…	be able to view a menu selections summary
In order to…	place an order

USER STORY 7	
Story Title	Pay Bill
Story ID	007
Story Type	User
Functional Area	Design and Structure Order Summary Receipt
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	finished dining
I want to…	be able to review all menu selections
In order to…	pay the bill

USER STORY 8	
Story Title	Leave Tip
Story ID	008
Story Type	User
Functional Area	Design and Structure Receipt
Non-Functional Area	Usability Performance
Priority	1
As a…	Customer
When…	paying the bill
I want to…	be able to add to the total
In order to…	leave a tip

USER STORY 9	
Story Title	Adjust Sales Tax
Story ID	009
Story Type	User
Functional Area	Design and Structure Receipt Administration
Non-Functional Area	Usability Performance
Priority	1
As a…	Administrator
When…	administering the OMS
I want to…	be able to adjust the sales tax
In order to…	comply with state sales tax regulations

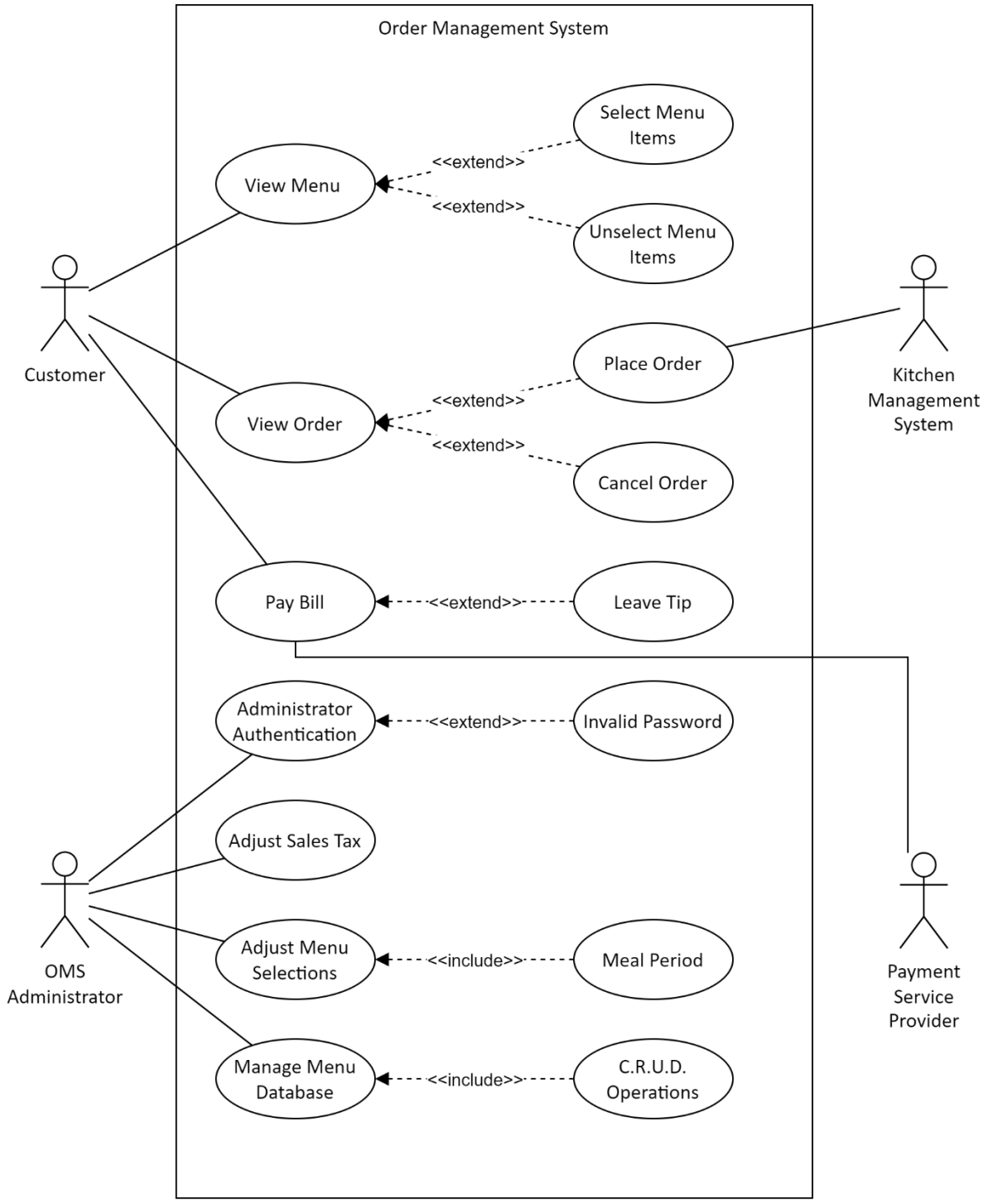
USER STORY 10	
Story Title	Adjust Menu Selections
Story ID	010
Story Type	User
Functional Area	Design and Structure Menu Items and Selection Administration
Non-Functional Area	Usability Performance
Priority	1
As a…	Administrator
When…	administering the OMS
I want to…	be able to turn menu items on and off
In order to…	adjust the menu based on the meal period

USER STORY 11	
Story Title	Manage Menu Database
Story ID	011
Story Type	User
Functional Area	Design and Structure Menu Items and Selection Administration
Non-Functional Area	Usability Performance
Priority	1
As a…	Administrator
When…	administering the OMS
I want to…	be able to manage the menu items database
In order to…	perform C.R.U.D. operations

USER STORY 12	
Story Title	Administrator Authentication
Story ID	012
Story Type	User
Functional Area	Design and Structure Administration
Non-Functional Area	Usability Performance Security
Priority	1
As a…	Administrator
When…	administering the OMS
I want to…	be able to control access to administrator utilities
In order to…	protect the integrity of the menu items database

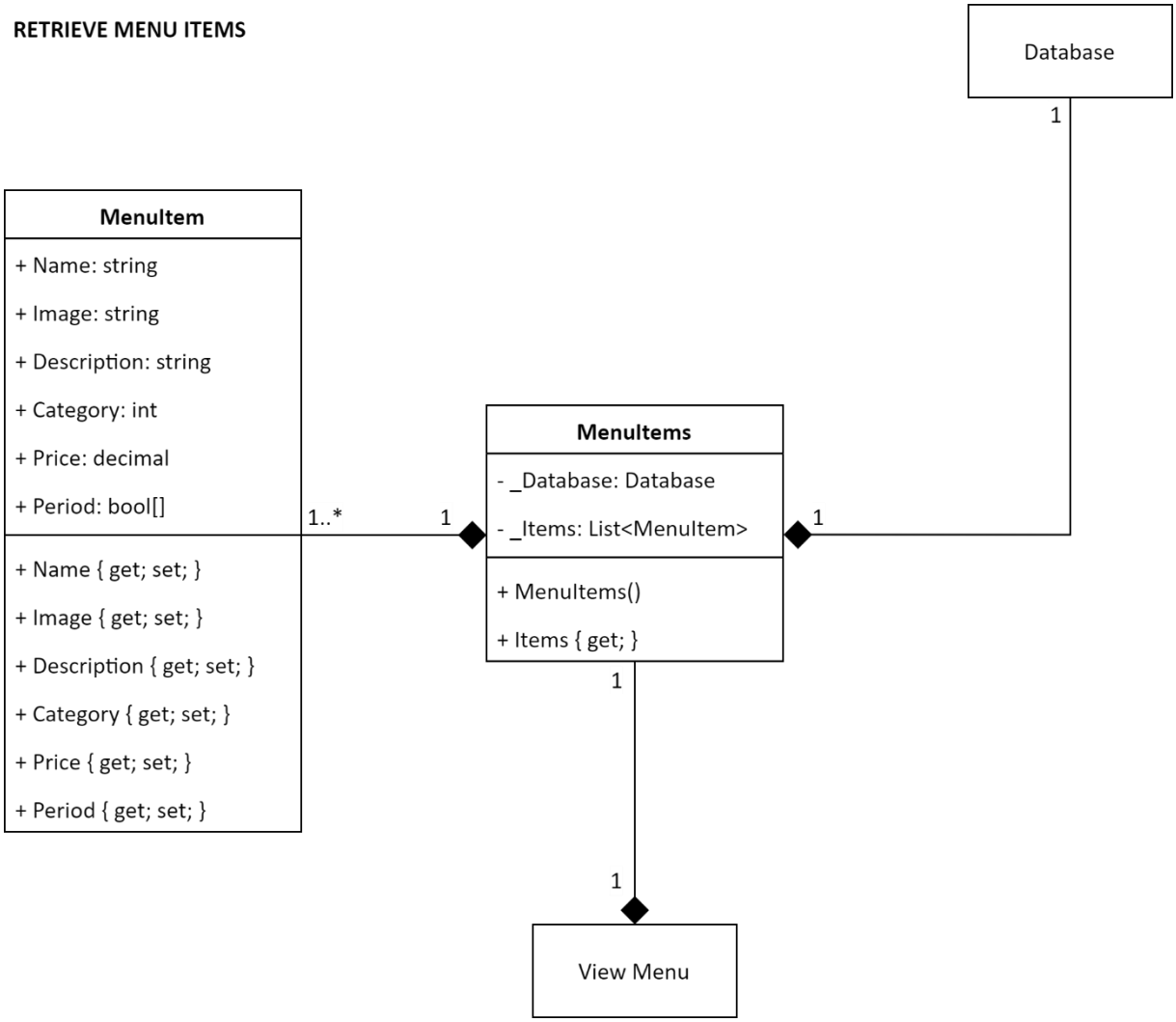
USE-CASE DIAGRAM

The Willow Tree OMS USE-CASE

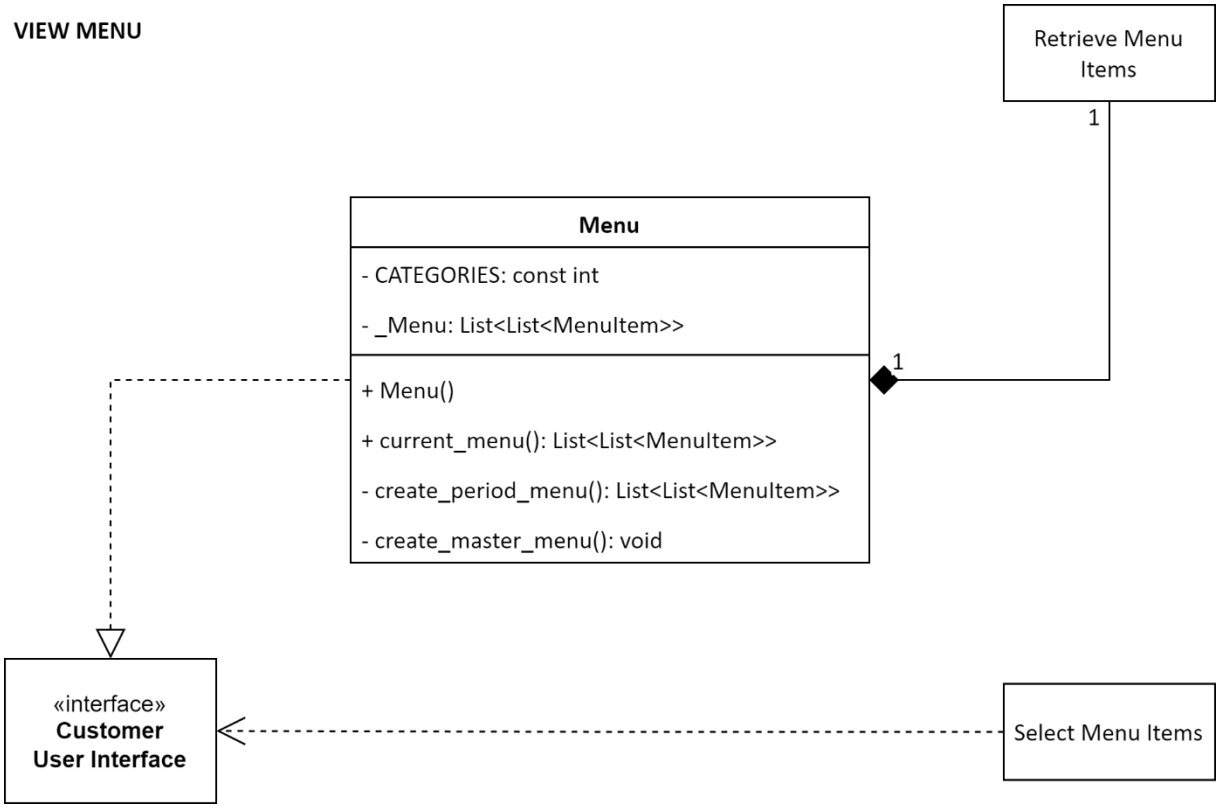


CLASS DIAGRAMS

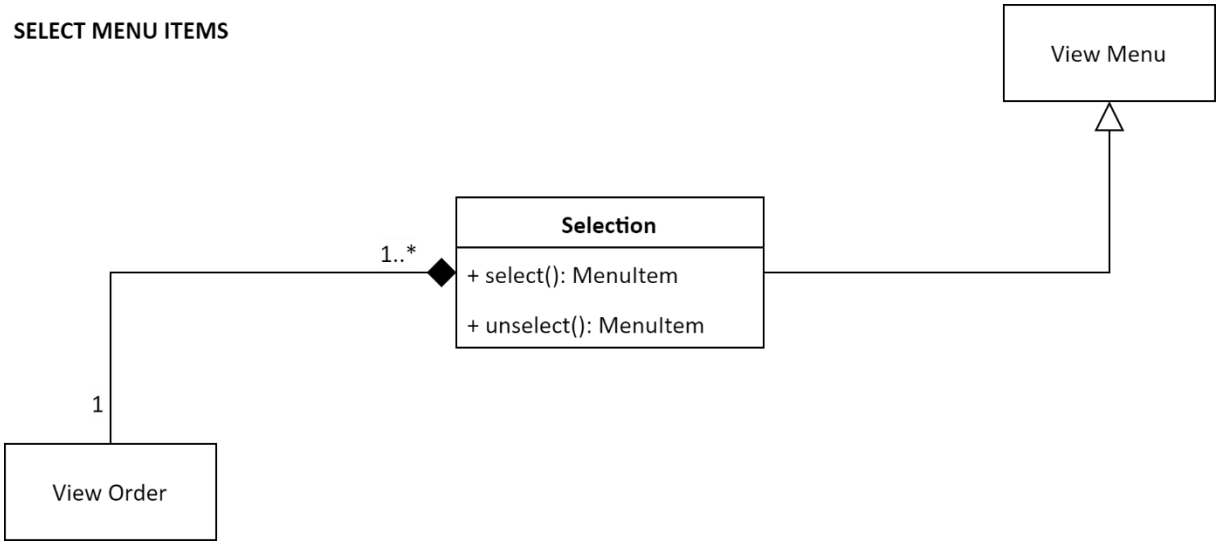
RETRIEVE MENU ITEMS



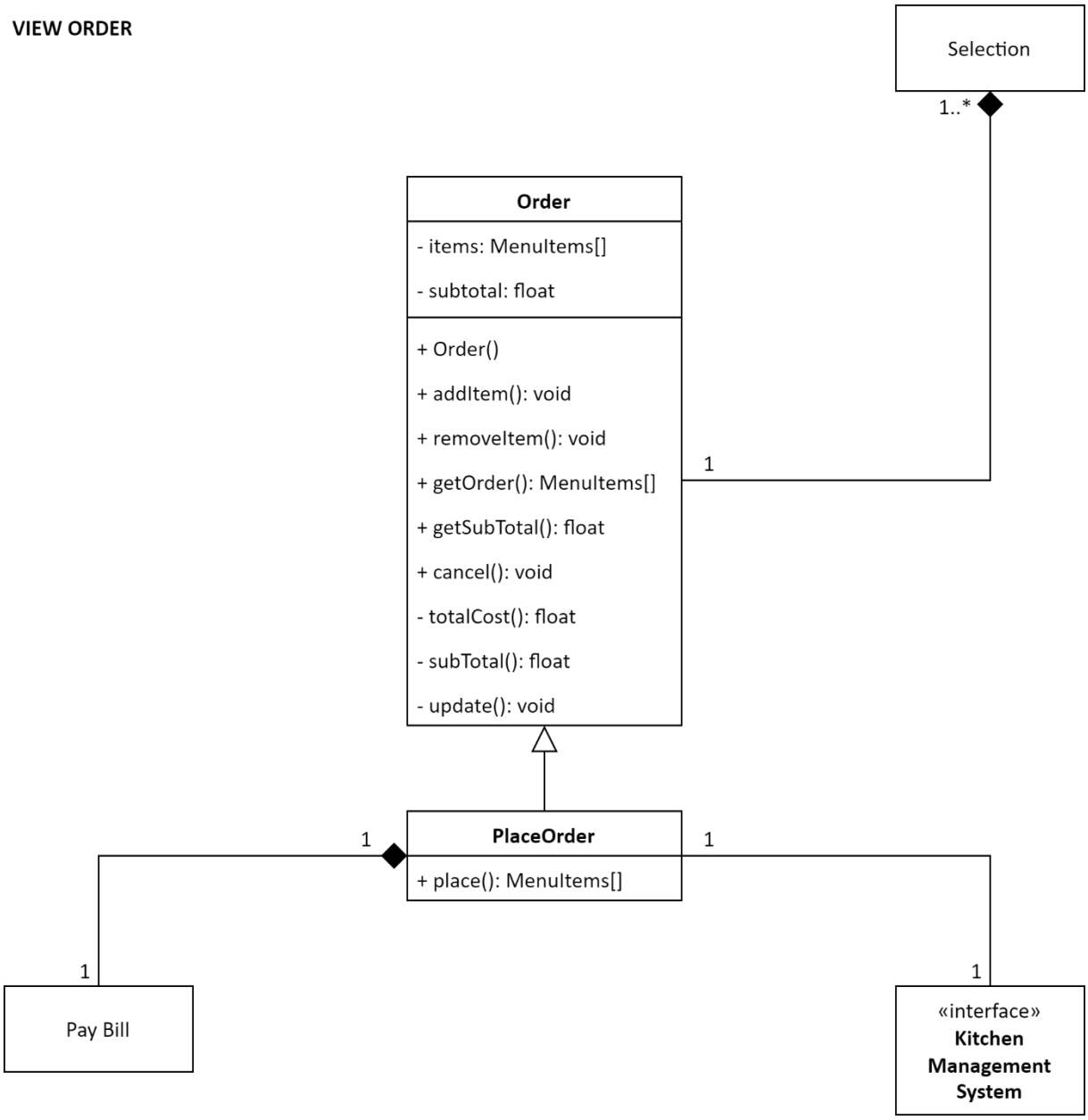
VIEW MENU


Retrieve Menu Items
1
Menu
- CATEGORIES: const int
- _Menu: List<List<MenuItem>>
+ Menu()
+ current_menu(): List<List<MenuItem>>
- create_period_menu(): List<List<MenuItem>>
- create_master_menu(): void
1
«interface»
Customer
User Interface
Select Menu Items

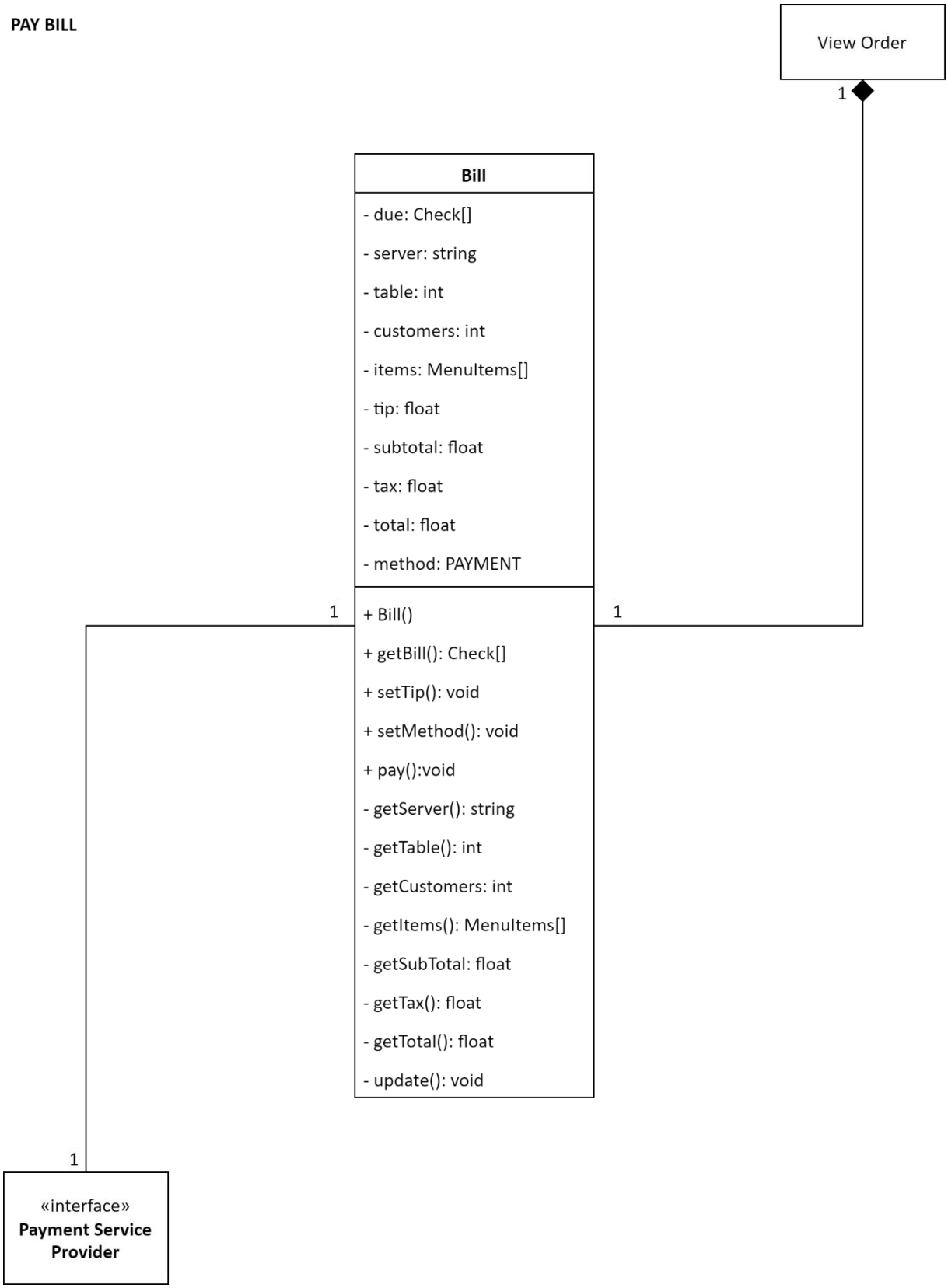
SELECT MENU ITEMS



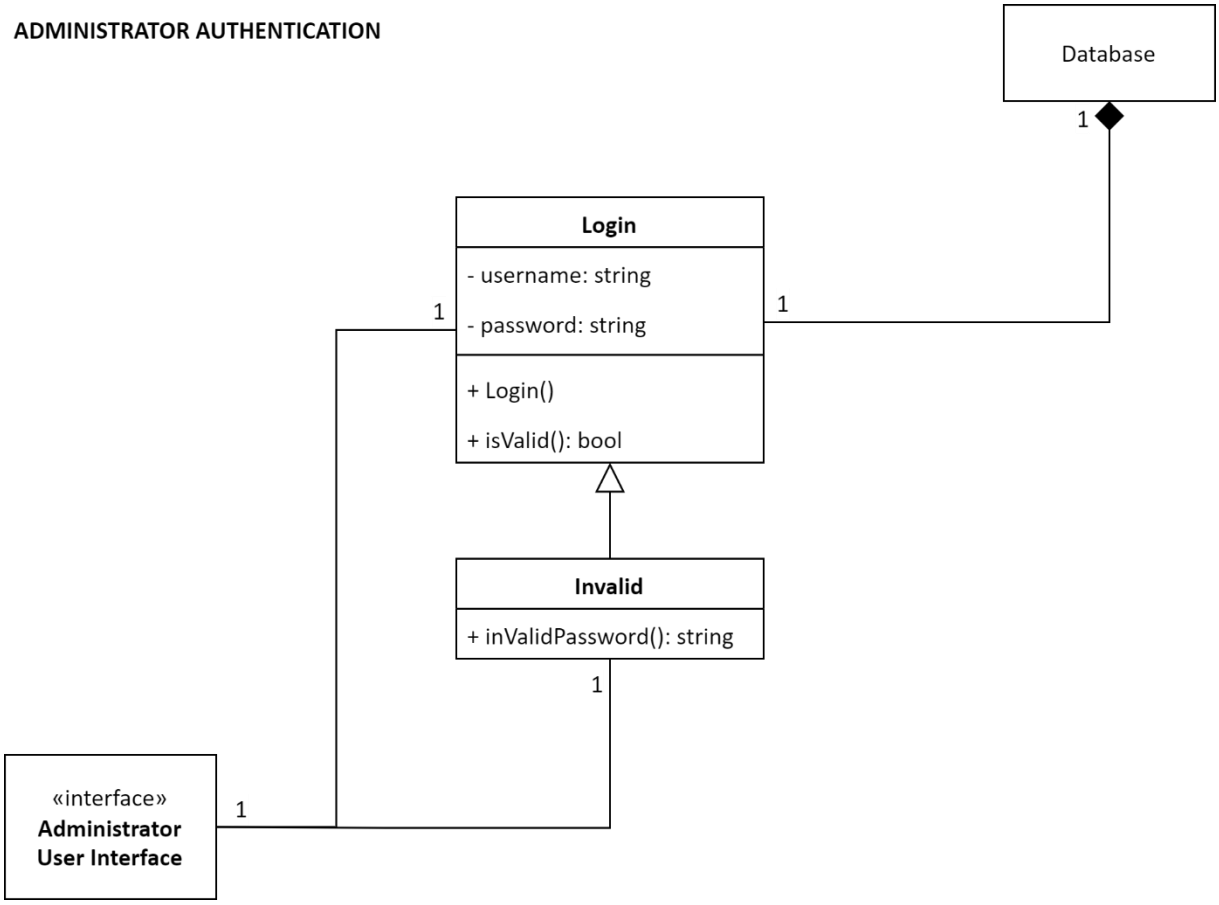
VIEW ORDER


Selection
1..*
Order
- items: MenuItems[]
- subtotal: float
+ Order()
+ addItem(): void
+ removeItem(): void
+ getOrder(): MenuItems[]
+ getSubTotal(): float
+ cancel(): void
- totalCost(): float
- subTotal(): float
- update(): void
1
PlaceOrder
+ place(): MenuItems[]
1
1
1
1
Pay Bill
«interface»
Kitchen
Management
System

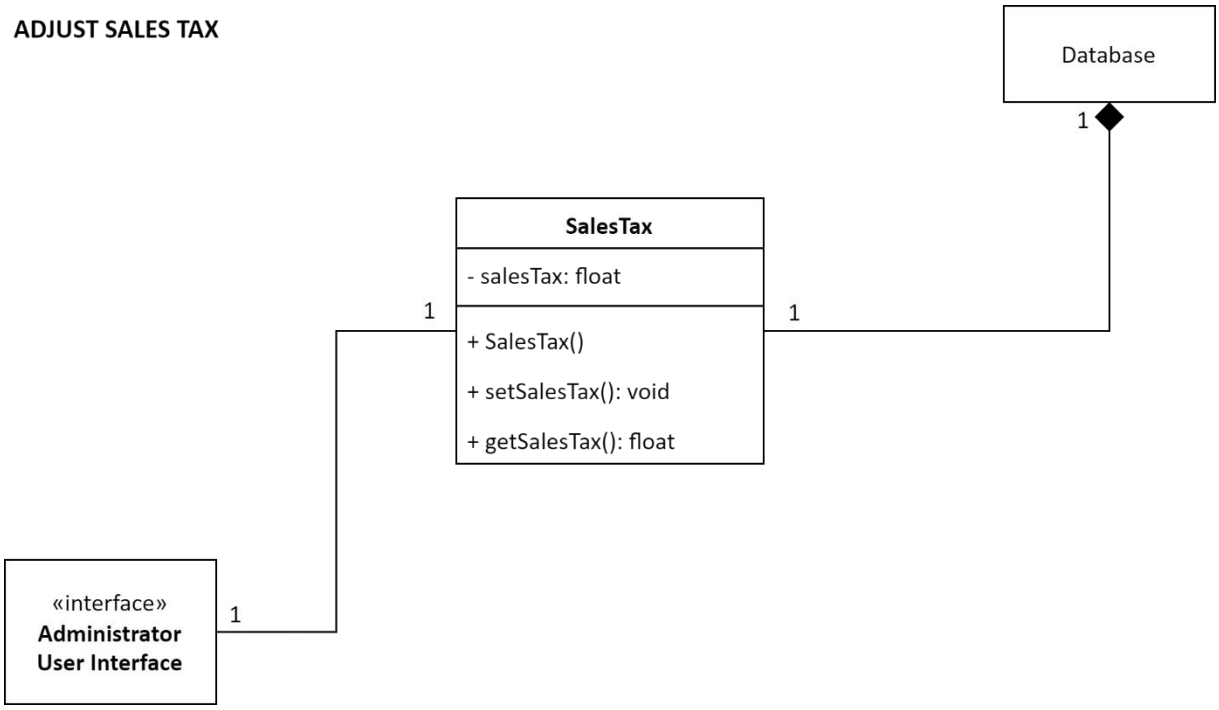
PAY BILL


View Order
1
Bill
- due: Check[]
- server: string
- table: int
- customers: int
- items: MenuItems[]
- tip: float
- subtotal: float
- tax: float
- total: float
- method: PAYMENT
+ Bill()
+ getBill(): Check[]
+ setTip(): void
+ setMethod(): void
+ pay():void
- getServer(): string
- getTable(): int
- getCustomers: int
- getItems(): MenuItems[]
- getSubTotal: float
- getTax(): float
- getTotal(): float
- update(): void
1
1
1
«interface»
Payment Service Provider

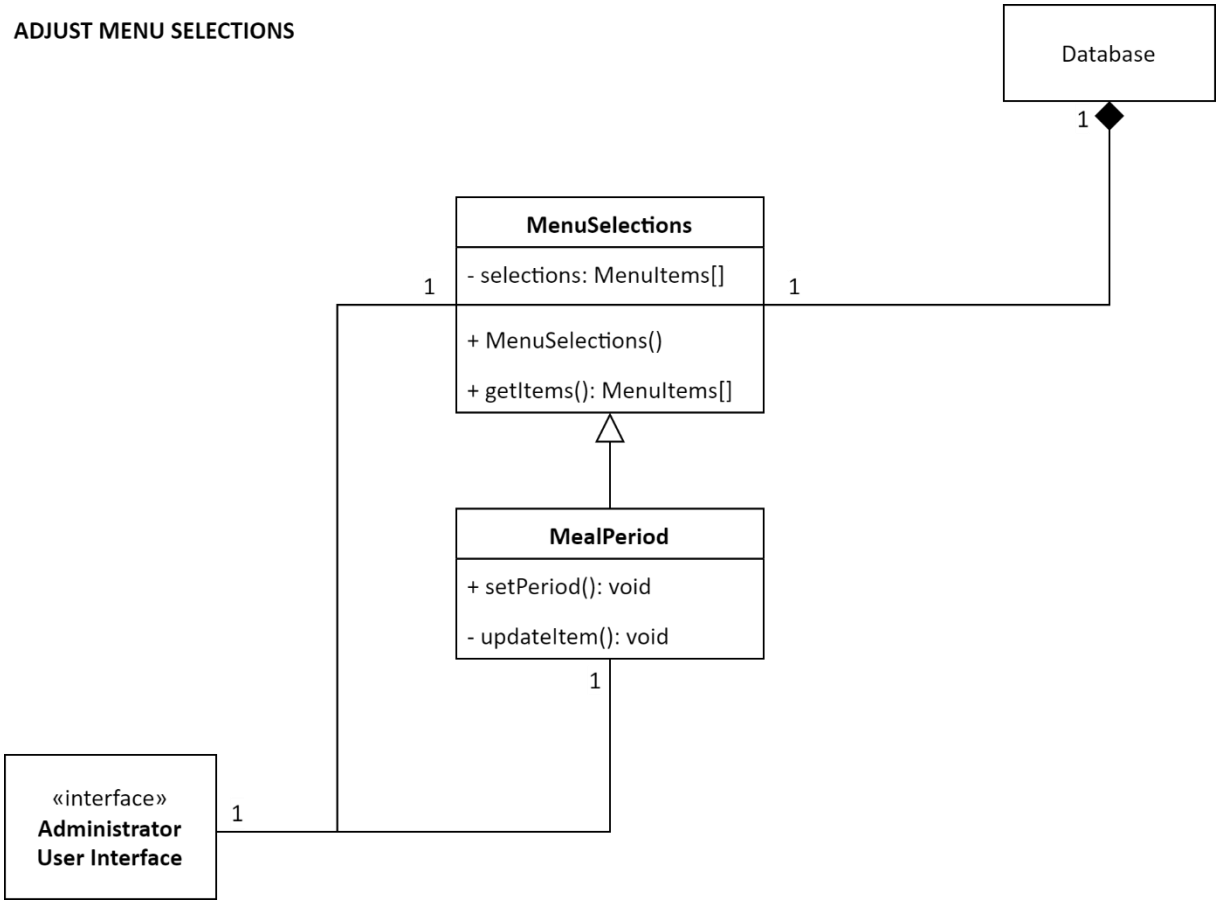
ADMINISTRATOR AUTHENTICATION



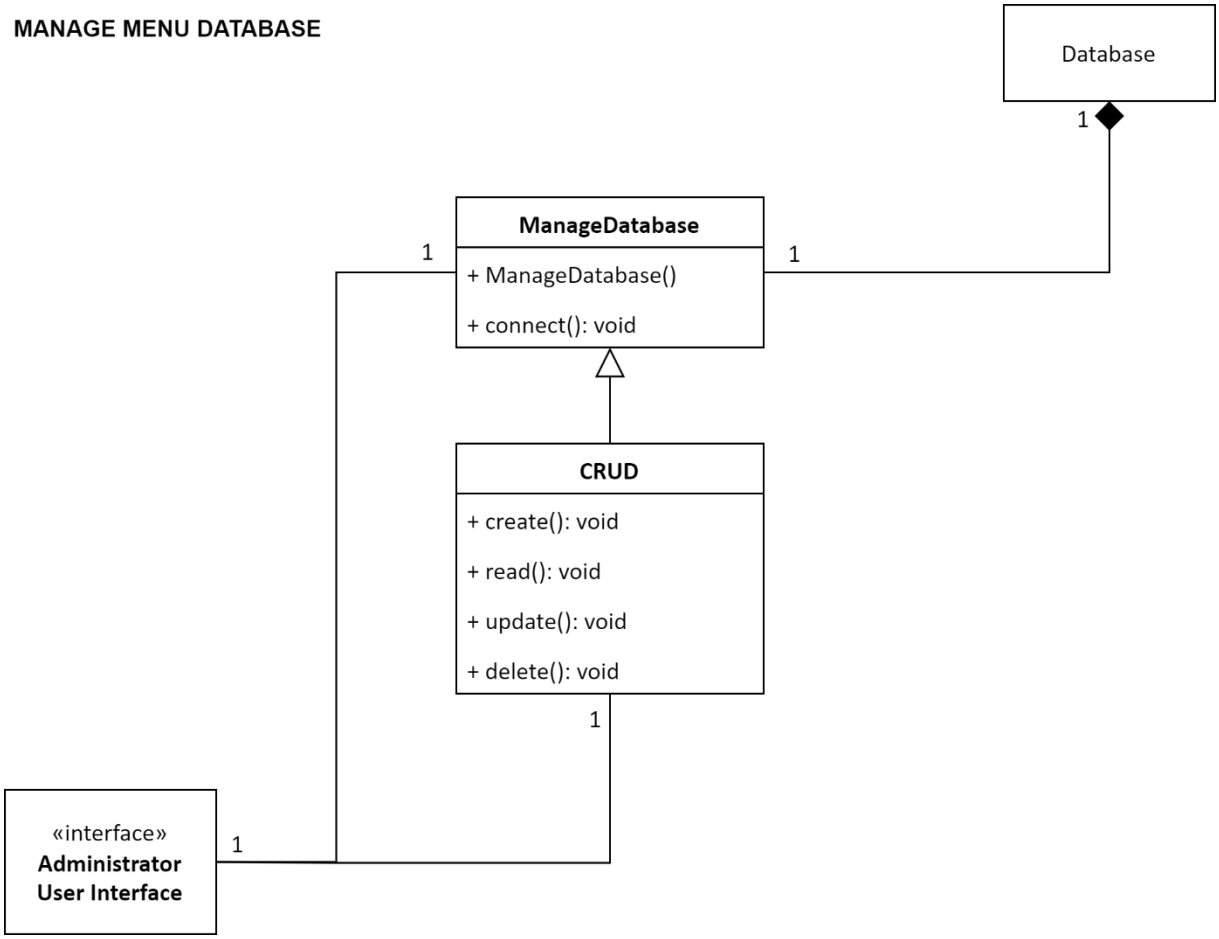
ADJUST SALES TAX



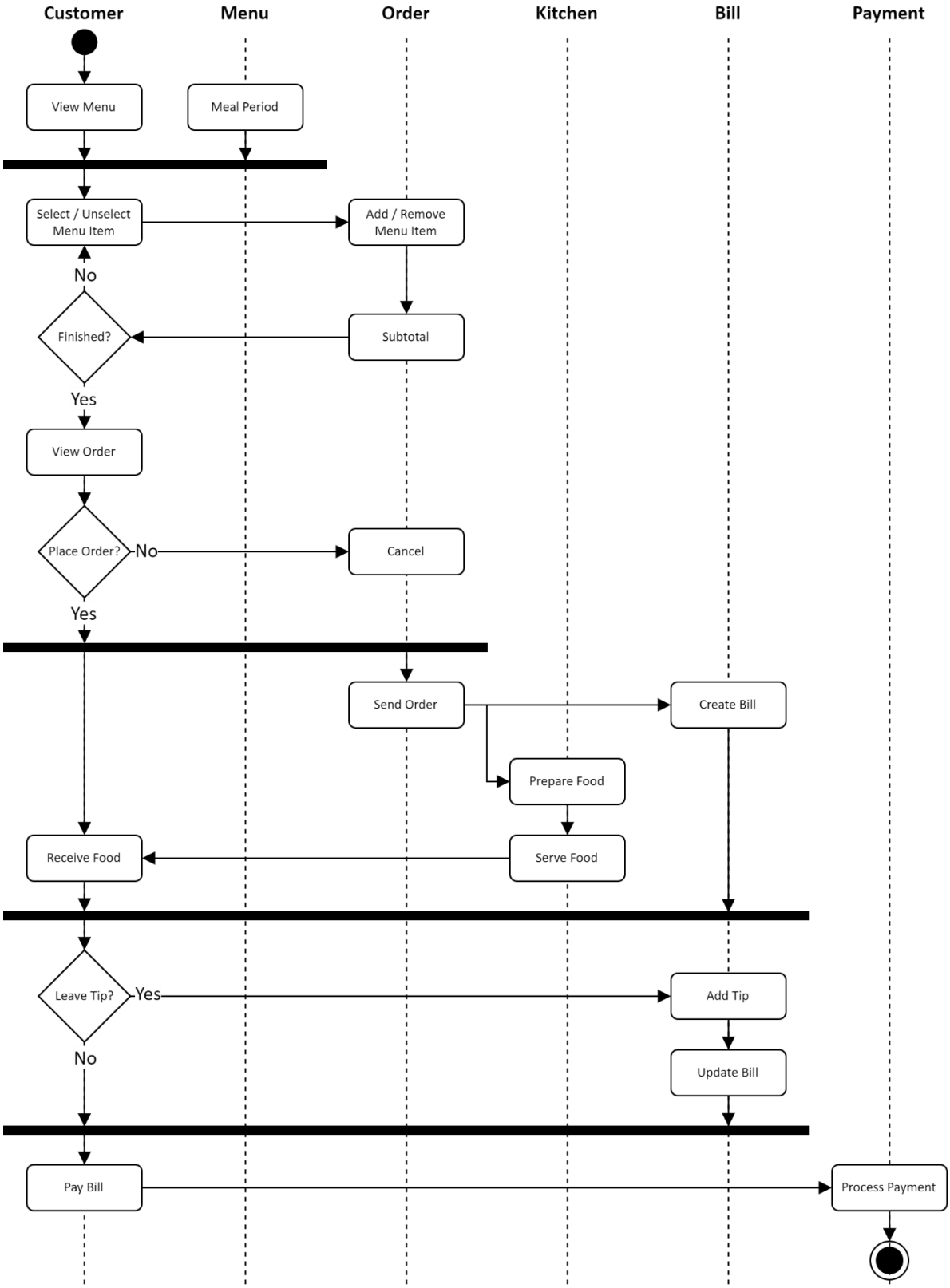
ADJUST MENU SELECTIONS

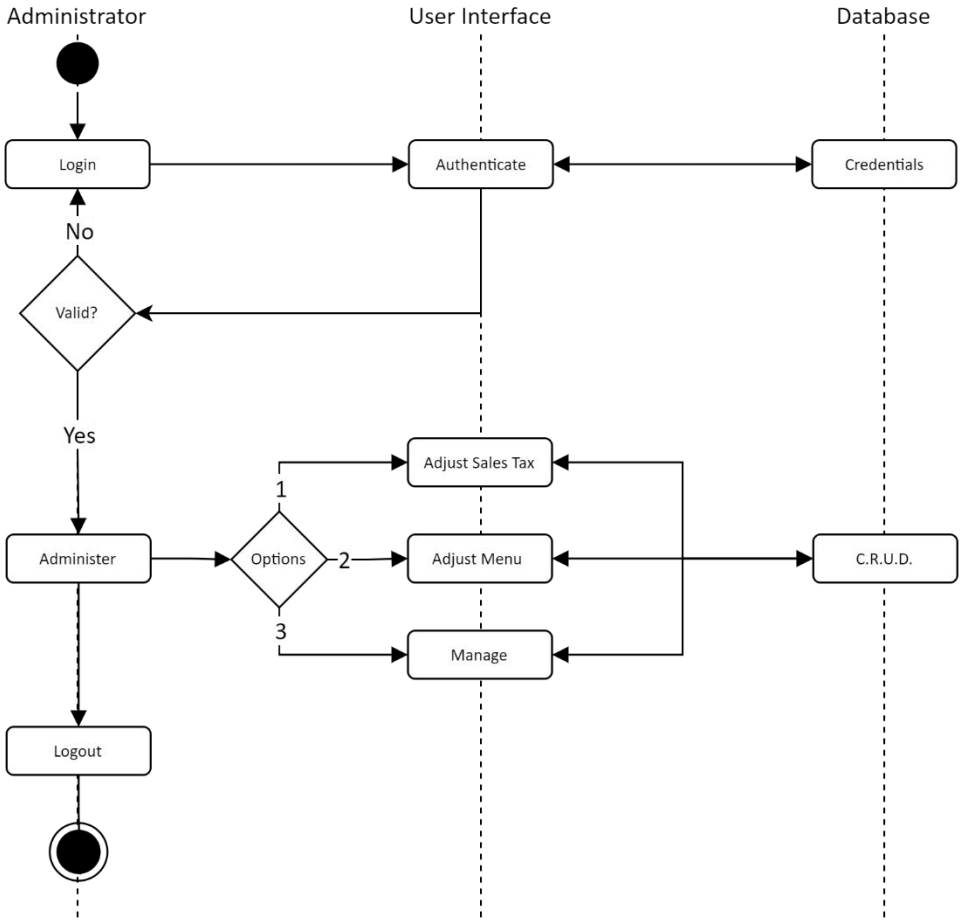


MANAGE MENU DATABASE

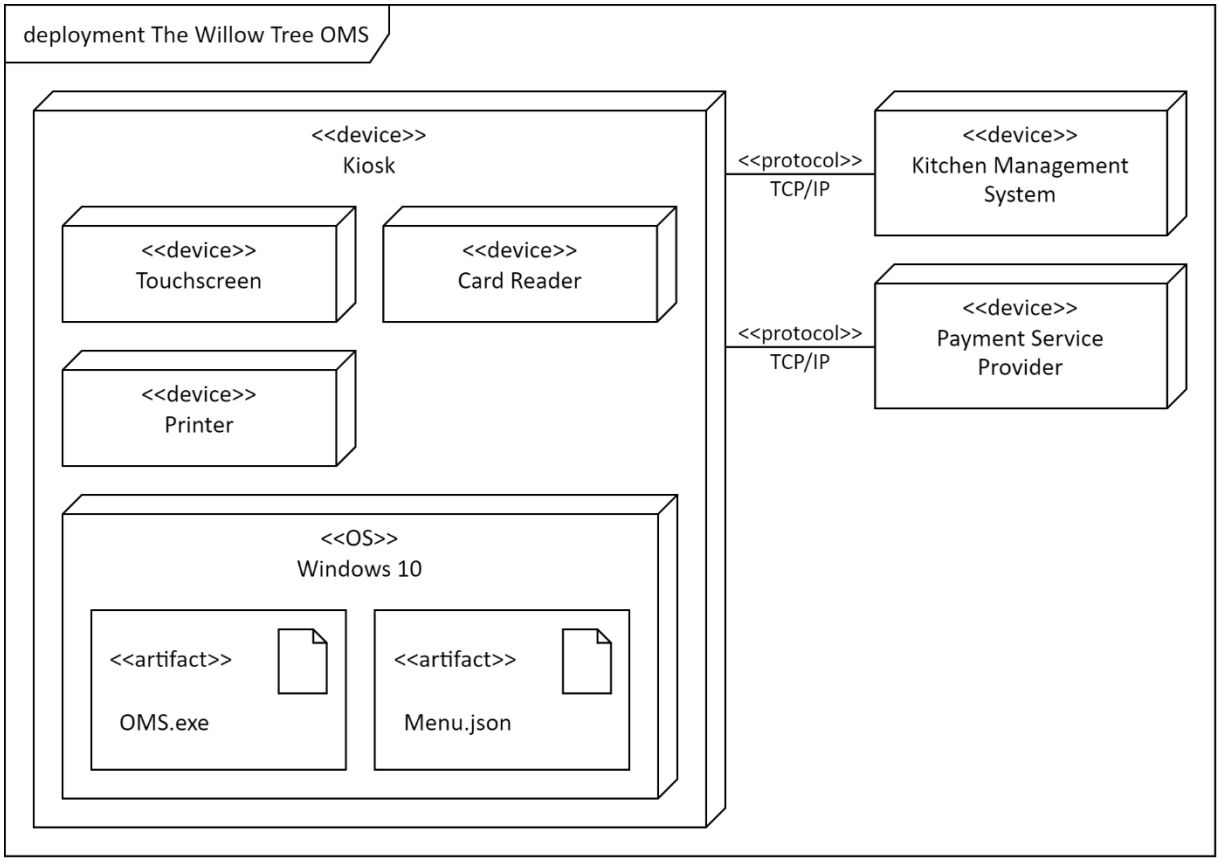


ACTIVITY DIAGRAMS




Administrator
User Interface
Database
Login
Authenticate
Credentials
No
Valid?
Yes
Adjust Sales Tax
1
Administer
Options
2
Adjust Menu
C.R.U.D.
3
Manage
Logout

DEPLOYMENT DIAGRAM



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